

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

```
In [2]: from sklearn.datasets import fetch_california_housing

housing = fetch_california_housing()
```

```
In [3]: df = pd.DataFrame(housing.data, columns=housing.feature_names)
df['Price'] = housing.target
df.head()
```

```
Out[3]:
```

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	Price
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23	4.526
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22	3.585
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	-122.24	3.521
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	-122.25	3.413
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	-122.25	3.422

```
In [4]: df.shape
```

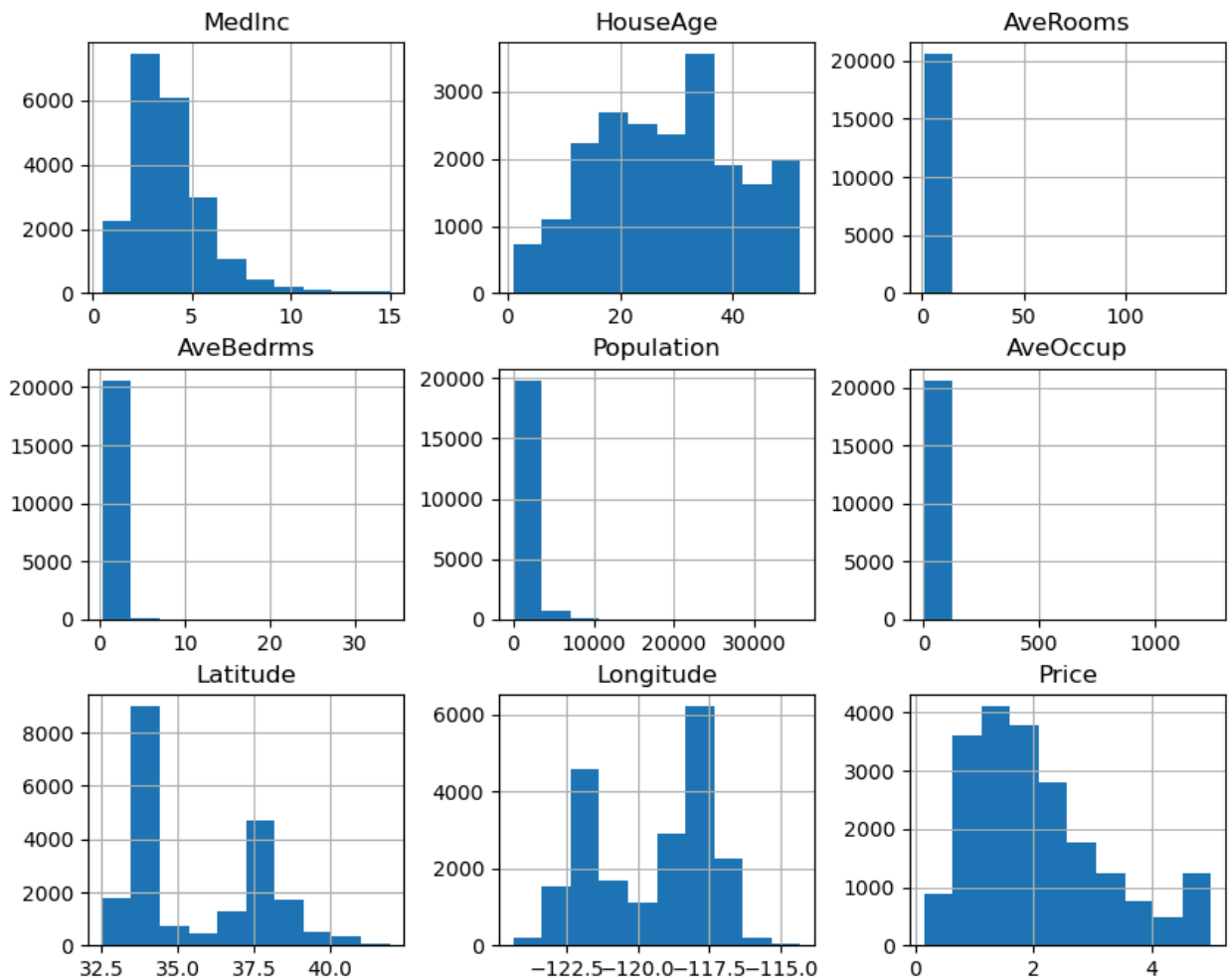
```
Out[4]: (20640, 9)
```

```
In [5]: df.describe()
```

```
Out[5]:
```

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude
count	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000
mean	3.870671	28.639486	5.429000	1.096675	1425.476744	3.070655	35.631861	-122.259984
std	1.899822	12.585558	2.474173	0.473911	1132.462122	10.386050	2.135952	1.923279
min	0.499900	1.000000	0.846154	0.333333	3.000000	0.692308	32.540000	-122.500000
25%	2.563400	18.000000	4.440716	1.006079	787.000000	2.429741	33.930000	-122.430000
50%	3.534800	29.000000	5.229129	1.048780	1166.000000	2.818116	34.260000	-122.390000
75%	4.743250	37.000000	6.052381	1.099526	1725.000000	3.282261	37.710000	-122.300000
max	15.000100	52.000000	141.909091	34.066667	35682.000000	1243.333333	41.950000	-122.100000

```
In [6]: df.hist(figsize=(10,8))
plt.show()
```



```
In [7]: X = df.drop('Price', axis=1)
        y = df['Price']
```

```
In [8]: X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=42
    )
```

```
In [9]: model = LinearRegression()
        model.fit(X_train, y_train)
```

```
Out[9]: LinearRegression ⓘ ⓘ
        Parameters
```

```
In [10]: y_pred = model.predict(X_test)
```

```
In [11]: mse = mean_squared_error(y_test, y_pred)
        print("Mean squared error:", mse)
```

Mean squared error: 0.5558915986952444

```
In [12]: r2 = r2_score(y_test, y_pred)
        print("R2 score:", r2)
```

R2 score: 0.5757877060324508

```
In [13]: sample_house = [X.iloc[0]]
        model.predict(sample_house)
```

```
C:\Users\ashish kumar\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
warnings.warn(
```

```
Out[13]: array([4.15194269])
```

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In [ ]:
```